

王羽丞 Adam

第二章 函数的导数(Derivative)

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2.1 导数的定义(The Definition of Derivative)



Multiple Choice Questions

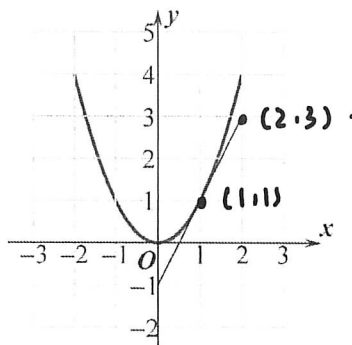
C 1. Which of the following values is the average rate of change of $f(x) = \sqrt{x+2}$ over the interval $[-1, 2]$?

- (A) -4 (B) $-\frac{1}{3}$ (C) $\frac{1}{3}$ (D) 3
- Handwritten calculation: $\frac{4 - \sqrt{1}}{4 - 3} = \frac{3}{1} = 3$*

C 2. Let $f(x) = 4 - 3x$. Which of the following is equal to $f'(0)$?

- (A) 4 (B) -4 (C) -3 (D) 4

3. The graph of f and its tangent line at $x=1$ is shown below, find $f'(1) =$ (D).



- (A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2

4. Let $f(x) = kx^3 - 2x$, ($k \neq 0$). The average rate of $f(x)$ over the interval $[-2, -1]$ is 3, find k .

Handwritten calculation: $\frac{4 - 2k + k - 2}{-3} = 3 \Rightarrow \frac{2 - k}{-3} = 3 \Rightarrow 2 - k = -9 \Rightarrow k = 11$

- (A) $\frac{5}{7}$ (B) $\frac{1}{7}$ (C) $-\frac{1}{7}$ (D) $-\frac{5}{7}$

C 5. **Calculator** Let $y = \tan e^x - x^2 + 1$. Find $\frac{dy}{dx} \big|_{x=0.5}$

- (A) -12.056 (B) 0 (C) 271.066 (D) 273.066

C 6. **Calculator** The graph of $y = \sqrt{x^2 + 1} - 3x$ crosses the x -axis at one point in the interval $[0, 1]$. What is the slope of the graph at this point?

- (A) 0.354 (B) -0.001 (C) -2.666 (D) 0

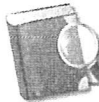
7. The population of insects at time t days is modeled by the function P with first derivative $P'(t) = 0.1t^2 + 10t + 120$. Which of the following statements is true?

B

- (A) At time $t = 10$, the population of insects is 230.
 (B) At time $t = 10$, the population of insects is increasing at a rate of 230 insects per day.
 (C) At time $t = 10$, the population of insects is decreasing at a rate of 230 insects per day.
 (D) At time $t = 10$, the rate of change of the number of insects is increasing at a rate of 230 insects per day.

8. If the line tangent to the graph of a function g at the point $(1, 4)$ passes through the point $(-2, -5)$, what is the slope of this line?

- (A) 3 (B) -3 (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$



Free Response Questions

9. Let $f(x) = x^2 - 2x$

(a) Find the average rate of change of f over $[1, 3]$.

(b) Find $\frac{f(2+h) - f(2)}{h}$.

(c) Find the instantaneous rate of change of f at $x = 2$.

$$\begin{aligned}
 f(1) &= -1 & f(3) &= 3 \\
 \frac{f(3) - f(1)}{3 - 1} &= \frac{3 - (-1)}{3 - 1} = \frac{4}{2} = 2 \\
 \frac{f(2+h) - f(2)}{h} &= \frac{(2+h)^2 - 2(2+h) - (2^2 - 2 \cdot 2)}{h} \\
 &= \frac{4 + 4h + h^2 - 4 - 2h - 4 + 4}{h} = \frac{h^2 + 2h}{h} = h + 2
 \end{aligned}$$

10. Use the definition of derivative to find the derivative of $f(x) = x^3$, then find $f'(2)$.

$$\begin{aligned}
 \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} &= \lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x)^3 - x^3}{\Delta x} \\
 &= \lim_{\Delta x \rightarrow 0} \frac{x^3 + 3x^2\Delta x + 3x\Delta x^2 + \Delta x^3 - x^3}{\Delta x} \\
 &= \lim_{\Delta x \rightarrow 0} (3x^2 + 3x\Delta x + \Delta x^2) \\
 &= 3x^2 \\
 f'(2) &= 3 \cdot 2^2 = 12
 \end{aligned}$$